Project 1 Pales in Comparison
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STQM 322
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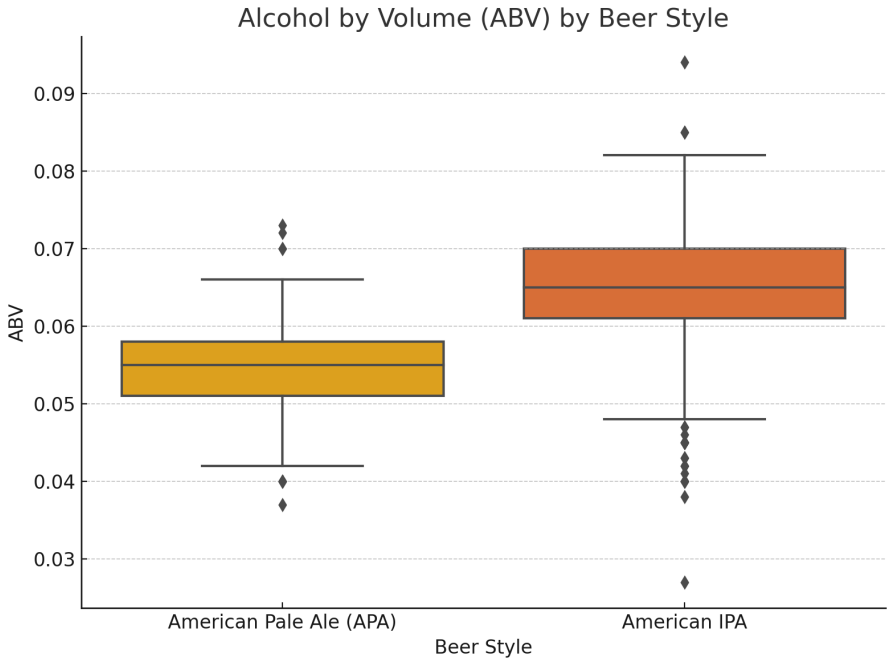
1. (1) (Explanation) I began by cleaning the dataset in Excel, removing the first two rows that contained metadata so the headers aligned properly. I then imported the cleaned CSV into Python using pandas for data manipulation. I filtered the data to include only 'American IPA' and 'American Pale Ale (APA)', dropped missing values, and converted ABV values to numeric.

   I calculated summary statistics (mean, standard deviation, min, max, quartiles) grouped by beer style. To compare the means, I used an independent samples t-test from the scipy.stats library, which is appropriate for comparing two independent groups. I also visualized the ABV distributions using a boxplot created with matplotlib and seaborn to show differences in alcohol content between the two beer types.

   (Boxplot) I compared the alcohol by volume (ABV) of American IPAs and American Pale Ales: I was able to recognize that IPAs generally have higher ABV, as well there's more variability in IPA alcohol levels.



(Summary Statistics)

Result

	count	mean	std	min	25%	50%
beer style						
American IPA	424.0	0.064503	0.008809	0.027	0.061	0.065
American Pale Ale (APA)	245.0	0.054686	0.006350	0.037	0.051	0.055

	75%	max
beer style		
American IPA	0.070	0.094
American Pale Ale (APA)	0.058	0.073

,

(16.6510875942366, 6.2685294121377e-52))

(T-Test Result)

t-statistic: 16.65
p-value: 6.27×10^{-52} (extremely small)

The p-value is far below 0.05, so we reject the null hypothesis. There is a statistically significant difference between the mean ABV of American IPAs and American Pale Ales.

(Python Code)

```
# Set the first row as the actual column headers

df_cleaned.columns = df_cleaned.iloc[0]
```

```
df_cleaned = df_cleaned.drop(index=0).reset_index(drop=True)

# Now reselect only the relevant columns for analysis

df_filtered = df_cleaned[['beer style', 'abv']].copy()

# Drop missing values

df_filtered.dropna(inplace=True)

# Convert ABV to numeric

df_filtered['abv'] = pd.to_numeric(df_filtered['abv'], errors='coerce')

df_filtered.dropna(subset=['abv'], inplace=True)

# Filter for American IPA and American Pale Ale

df_filtered = df_filtered[df_filtered['beer style'].isin(['American IPA', 'American
Pale Ale (APA)'])]

# Summary statistics

summary_stats = df_filtered.groupby('beer style')['abv'].describe()

# Perform t-test

from scipy.stats import ttest_ind

ipa_abv = df_filtered[df_filtered['beer style'] == 'American IPA']['abv']

pale_ale_abv = df_filtered[df_filtered['beer style'] == 'American Pale Ale
(APA)']['abv']

t_stat, p_value = ttest_ind(ipa_abv, pale_ale_abv, equal_var=False)

import matplotlib.pyplot as plt

import seaborn as sns

# Create a boxplot to visualize ABV by beer style

plt.figure(figsize=(8, 6))

sns.boxplot(data=df_filtered, x='beer style', y='abv')
```

```
plt.title('Alcohol by Volume (ABV) by Beer Style')

plt.ylabel('ABV')

plt.xlabel('Beer Style')

plt.tight_layout()

plt.show()

summary_stats, (t_stat, p_value)
```

(2)

(A) Null Hypothesis (H_0):
mIPA=mAPA - (The mean ABV of American IPAs is equal to the mean ABV of American Pale Ales.)

Alternative Hypothesis (H_1):
mIPA≠mAPA- (The mean ABV of American IPAs is not equal to the mean ABV of American Pale Ales.)

(B) Set at: $\alpha = 0.05$
(C) Independence: Beers are independent entries.
Normality: Sample sizes are large (n=424 IPA, n=245 APA), so the Central Limit Theorem applies.
Variances: Based on the boxplot and standard deviations, variances differ — use Welch's t-test.
(D) p-value: 6.27×10^{-52}
(E) Since $p < 0.05$, we reject the null hypothesis.
(F) There is strong evidence that the mean alcohol content differs between American IPAs and American Pale Ales. IPAs tend to have a significantly higher ABV.

(3)

Difference in means = 0.0098
Standard error ≈ 0.00059
t-critical ≈ 1.99 (from degrees of freedom)
Margin of error = ~ 0.0011

CI = (0.00866, 0.01098)

This means we are 95% confident that American IPAs have 0.87% to 1.10% higher ABV than American Pale Ales.

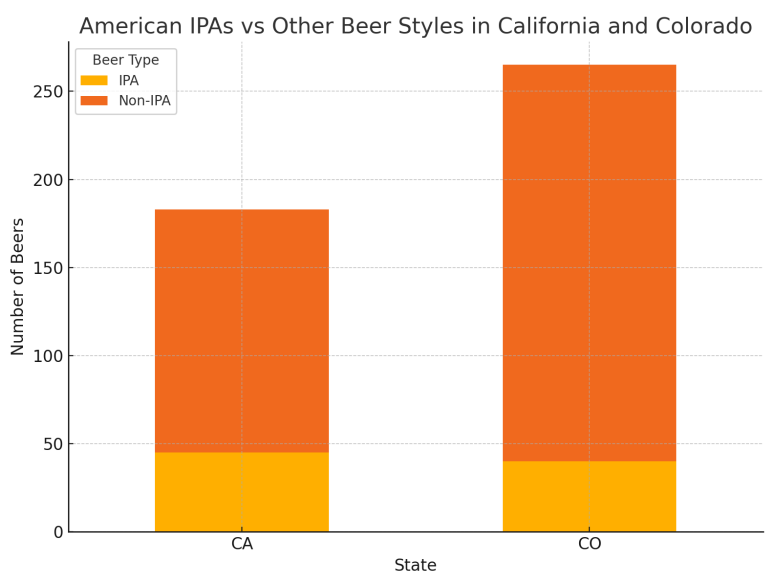
Interpretation of CI

Since the entire confidence interval is above 0, it supports our earlier conclusion from the hypothesis test — there is a real difference in alcohol content.

2.

(1) (Explanation) To compare the popularity of American IPAs in California and Colorado, I began by cleaning the dataset, removing any inconsistencies such as extra spaces in state names and filtering the data to include only beers from CA and CO. I created a new variable to label whether each beer was an American IPA or not. Using this, I grouped the data by state and beer type (IPA vs. non-IPA), and calculated how many beers of each type were produced in each state. I then computed the total number of beers and the proportion of American IPAs for both states. California showed that 24.6% of its beers were IPAs, while Colorado had 15.1%. I visualized these results with a stacked bar chart comparing the number of IPAs to non-IPAs in each state, which helped illustrate the higher concentration of IPAs in California's beer production.

(Bar Chart) The bar chart below compares the number of American IPAs to all other beer types in California and Colorado. In the chart I noticed California has a higher proportion of IPAs (≈ 24.6%) than Colorado (≈ 15.1%). This suggests that IPAs make up a larger share of the beer landscape in CA compared to CO.



(Summary Statistics)

	IPA	Non-IPA	Total	Proportion_IPA
CA	45	138	183	0.245902
CO	40	225	265	0.150943

(Two-Proportion Z-Test Result)

Z-statistic: 2.5197

P-value: 0.0117

Since the p-value (0.0117) is less than the standard significance level of 0.05, we reject the null hypothesis.

(Python Code)

```
import pandas as pd

import matplotlib.pyplot as plt

from statsmodels.stats.proportion import proportions_ztest

# Load the Excel file

excel_path = "Beers_and_Breweries.xlsx"  # Adjust the path if needed

xls = pd.ExcelFile(excel_path)

df = xls.parse('Beers_and_Breweries')

# Set proper headers and clean

df.columns = df.iloc[2]

df = df.drop(index=[0, 1, 2]).reset_index(drop=True)

# Select and clean relevant columns
```

```
df_cleaned = df[['beer style', 'abv', 'state']].copy()

df_cleaned.dropna(subset=['beer style', 'abv', 'state'], inplace=True)

df_cleaned['state'] = df_cleaned['state'].str.strip()


# Filter for California and Colorado

df_filtered_states = df_cleaned[df_cleaned['state'].isin(['CA', 'CO'])].copy()

df_filtered_states['is_ipa'] = df_filtered_states['beer style'] == 'American IPA'

# Group and count IPAs and non-IPAs

ipa_counts = df_filtered_states.groupby(['state',
'is_ipa']).size().unstack(fill_value=0)

ipa_counts.columns = ['Non-IPA', 'IPA']

ipa_counts['Total'] = ipa_counts.sum(axis=1)

ipa_counts['Proportion_IPA'] = ipa_counts['IPA'] / ipa_counts['Total']

# Plot stacked bar chart

ipa_counts[['IPA', 'Non-IPA']].plot(kind='bar', stacked=True, figsize=(8, 6))

plt.title('American IPAs vs Other Beer Styles in California and Colorado')

plt.ylabel('Number of Beers')

plt.xlabel('State')

plt.xticks(rotation=0)

plt.legend(title='Beer Type')

plt.tight_layout()

plt.show()

# Print summary statistics

print("\nSummary Statistics:")
```

```
print(ipa_counts[['IPA', 'Non-IPA', 'Total', 'Proportion_IPA']])

# Perform two-proportion z-test

ipa_successes = ipa_counts['IPA'].values

ipa_totals = ipa_counts['Total'].values

z_stat, p_val = proportions_ztest(count=ipa_successes, nobs=ipa_totals)

# Print test results

print("\nTwo-Proportion Z-Test Result:")

print(f"Z-statistic: {z_stat:.4f}")

print(f"P-value: {p_val:.4f}")

# Interpretation

if p_val < 0.05:

    print("Result: Reject the null hypothesis – the proportion of IPAs is significantly different.")

else:

    print("Result: Fail to reject the null hypothesis – no significant difference in proportions.")
```

(2)

(A) We are testing whether the proportion of American IPAs is the same in California and Colorado.

- Null Hypothesis (H_0):
  pCA=pCO (The proportion of American IPAs is the same in both states.)

- Alternative Hypothesis (H_1):
  pCA≠pCO (The proportion of American IPAs is different between California and Colorado.)

(B) Set at: $\alpha = 0.05$

(C) Assumptions for the two-proportion z-test: Independent samples: The beers from California and Colorado are separate and independently selected.
Categorical outcome: Each beer is either an "IPA" or "Not IPA" (binary).
Large sample size: Both groups have large enough sample sizes for normal approximation to apply:
CA: 45 IPAs, 138 non-IPAs → Total = 183
CO: 40 IPAs, 225 non-IPAs → Total = 265
→ Successes and failures in both groups are > 10 → assumption met.
Test Used: Two-proportion z-test
Test Statistic (from code): z = 2.5197

(D) p-value = 0.0117

(E) This is the probability of observing a difference this large (or larger) between the two proportions by random chance, assuming the null is true. Significance level: $\alpha = 0.05$, Since p-value (0.0117) < α, we reject the null hypothesis.

(F) (Conclusion) There is statistical evidence to conclude that the proportion of American IPAs is significantly different between California and Colorado. Specifically, California has a higher proportion of IPAs (24.6%) compared to Colorado (15.1%). This suggests that IPAs are more prevalent in the California beer market than in Colorado's.

(3) The 95% confidence interval for the difference in proportions (California − Colorado) is: (0.0191, 0.1708) This means we are 95% confident that the true difference in the proportion of American IPAs between California and Colorado lies between 1.91% and 17.08%. Since the entire confidence interval is greater than 0, it indicates that California has a higher proportion of IPAs than Colorado. The hypothesis test led us to reject the null hypothesis because the p-value (0.0117) was less than 0.05. The confidence interval does not include 0, which also implies that the difference is statistically significant. So, both the hypothesis test and the confidence interval agree: California has a significantly higher proportion of American IPAs compared to Colorado.